

CODEMIX SKILL

The safety + decision layer between AI agents and their actions

*“Order cancel **mat** karo, bas address change karna hai.”*



We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

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Initiative Overview

Initiative: Codemix Skill · Owner: Team Ramanathan & Sadhana · a plug-in that makes tool-using AI agents safe

The gap

Voice agents can understand a request and immediately call a refund, cancellation or replacement API. A small missed negation can turn the right conversation into the wrong transaction.

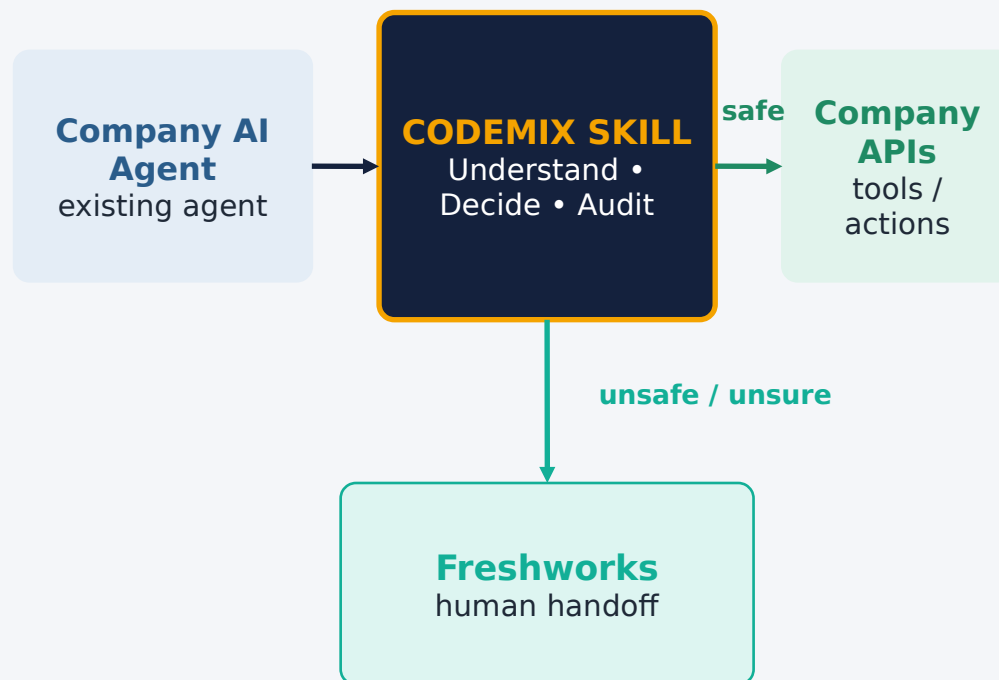
Our product

Codemix Skill sits between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

Freshworks extension

Unsafe, uncertain or emotionally escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

WHERE IT SITS



The agent proposes an action. Codemix Skill decides whether it runs, needs confirmation, or goes to a human.

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Problem Statement

In code-mixed voice, one short word can reverse the meaning of an API action

HINDI EXAMPLE

*Caller: "Order cancel **mat** karo, bas address change karna hai."*

Correct intent: do NOT cancel → update address.

TAMIL EXAMPLE

*Caller: "Refund **vendam**, replacement anuppunga."*

Correct intent: do NOT refund → request replacement.

Why this matters Speech-to-text can drop a negation on a noisy line, or an intent model can focus on words like "cancel" or "refund" and ignore the negation. With tool calling, that misunderstanding becomes a real transaction.

RISK

Wrong refund / cancellation

BUSINESS IMPACT

Money, reshipping, trust

NEED

A decision layer before action

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Why Now

AI agents have write access, so misunderstanding is no longer just a bad answer



AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations, returns and other business actions directly.



Code-mixing is normal

Indian callers switch between languages inside a sentence. The risk is concentrated around small intent-flipping words.



Voice stack is ready

VoBiz provides phone access + DTMF, Sarvam handles Indian speech-to-text, Claude validates intent, and ElevenLabs returns the response.

Our timing The missing layer is not another voice bot. It is a safety and decision layer that controls when an existing AI agent is allowed to call an API.

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Competitive Landscape

We focus specifically on the gap between language understanding and action execution

Do nothing

✓ Zero integration effort.

✗ Risky actions can execute on the agent's first interpretation.

Confirm-everything IVR

✓ Safe confirmation.

✗ Adds friction to every action, even low-risk requests.

Generic LLM guardrails

✓ Useful for policy, toxicity and PII.

✗ Not designed as an action gate for code-mixed negation + speech confidence.

Codemix Skill

Understands code-mixed intent



Checks action risk



ALLOW / CONFIRM / BLOCK



Routes unsafe cases to Freshworks

Friction on every action

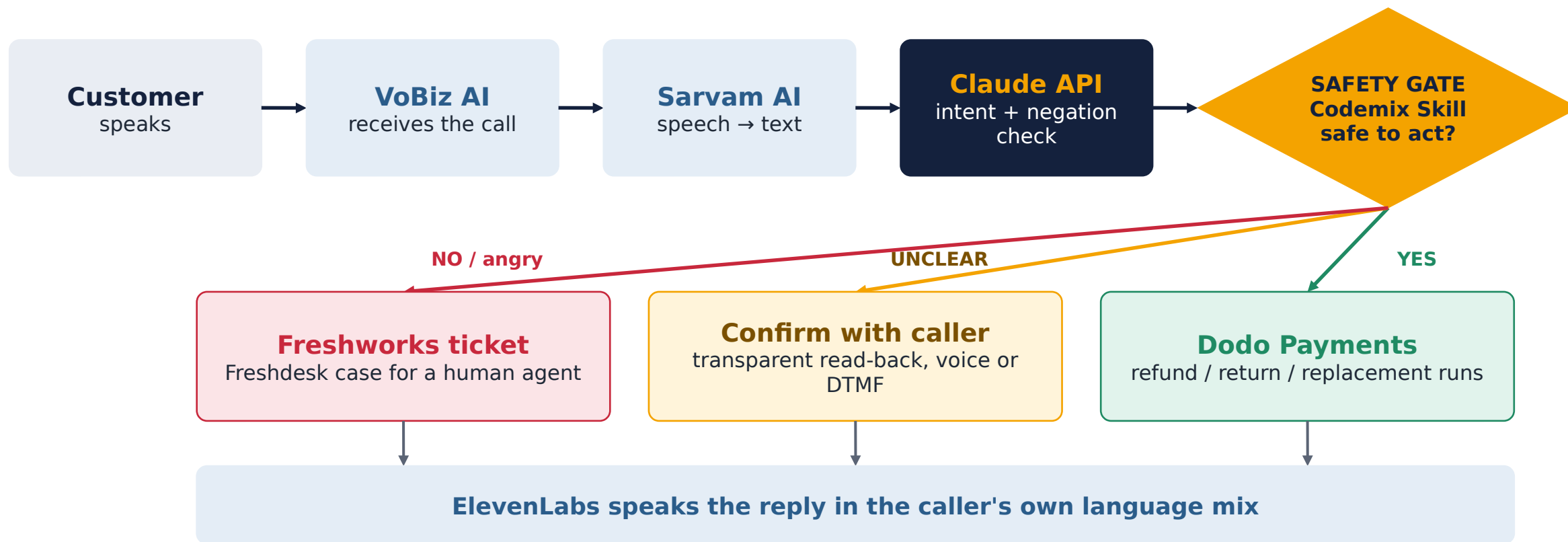
Risk-based: only risky actions get a check

Our edge A policy-aware action gate with human escalation and an auditable record of what the caller said, what the agent decided, and what API was called.

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Solution Architecture

Codemix Skill is the control point between an existing AI agent and its tools



Our innovation We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

Sponsor & Integration Stack

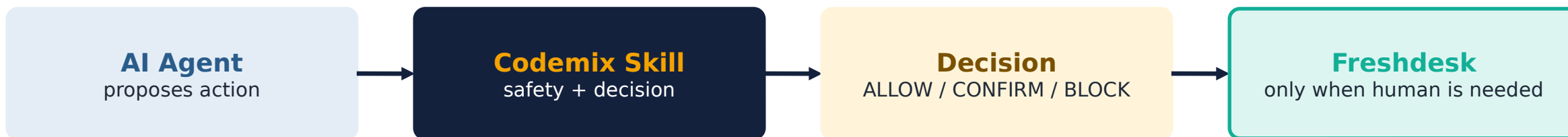
Each sponsor/API has one clear job; Codemix Skill coordinates the handoff and controls action execution

CHANNEL	VoBiz AI	PHONE + DTMF	The number customers call. Keypad (DTMF) gives a confirmation that needs no speech recognition.
LISTEN	Sarvam AI	SPEECH → TEXT	Transcribes code-mixed Hindi + English and Tamil + English, so the negation reaches the intent check.
VALIDATE	Claude API	INTENT + SAFETY	Before any sensitive action: finds the real intent. "Refund vendam" = NO refund, YES replacement.
CONTROL	Codemix Skill	ORCHESTRATION + SAFETY	The safety gate between what the voice agent hears and what it is allowed to do.
ACT	Dodo Payments	ACTION API	Runs refunds, returns and replacements, but only after the safety gate says YES.
ESCALATE	Freshworks / Freshdesk	HUMAN WORKFLOW	If it can't be resolved safely, a Freshdesk ticket hands the case to a human agent.
RESPOND	ElevenLabs	VOICE OUTPUT	Speaks the reply in the caller's own language mix and reads back what it understood.

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Freshworks Integration — Human-in-the-Loop

Freshdesk is the operational destination for cases the AI should not resolve alone



When a ticket is created

- Unsafe or unclear action
- Caller does not confirm
- Caller is angry / distressed
- AI should not guess



What Freshdesk receives

- English summary
- Full code-mixed transcript
- Intent + confidence
- Safety verdict
- Attempted API action
- Tags / timestamps



What the human gets

A ready-to-handle case instead of a blind escalation: context, reasoning and conversation history arrive with the ticket.

Codemix Skill

Freshdesk REST API

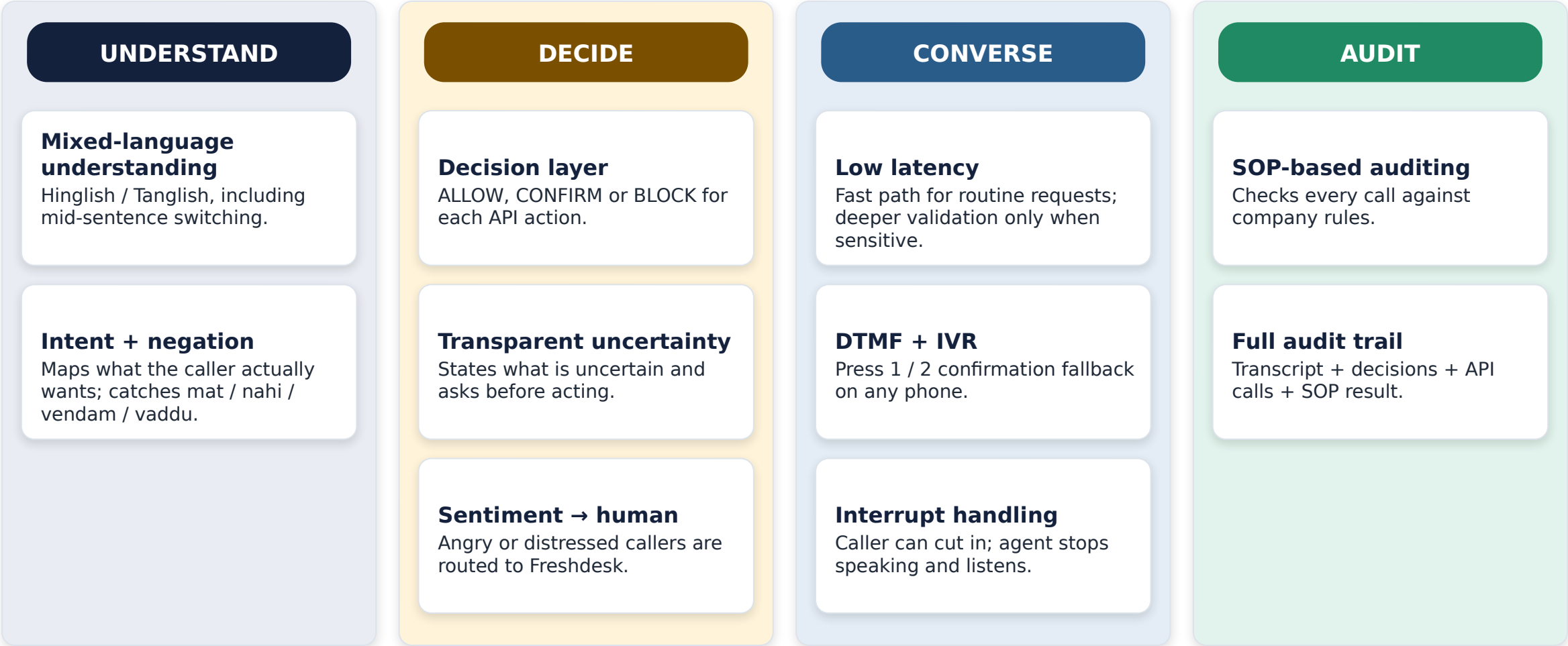
POST /api/v2/tickets

Human agent



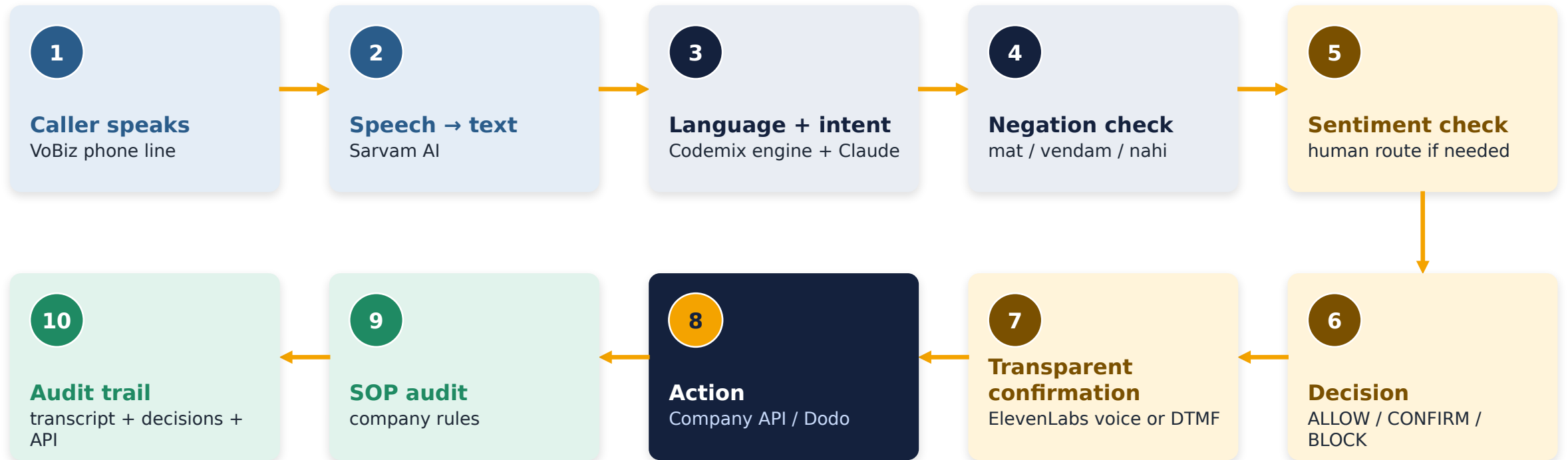
The Feature Set

The product combines understanding, decisioning, conversation control and auditability



10 One Call — End to End

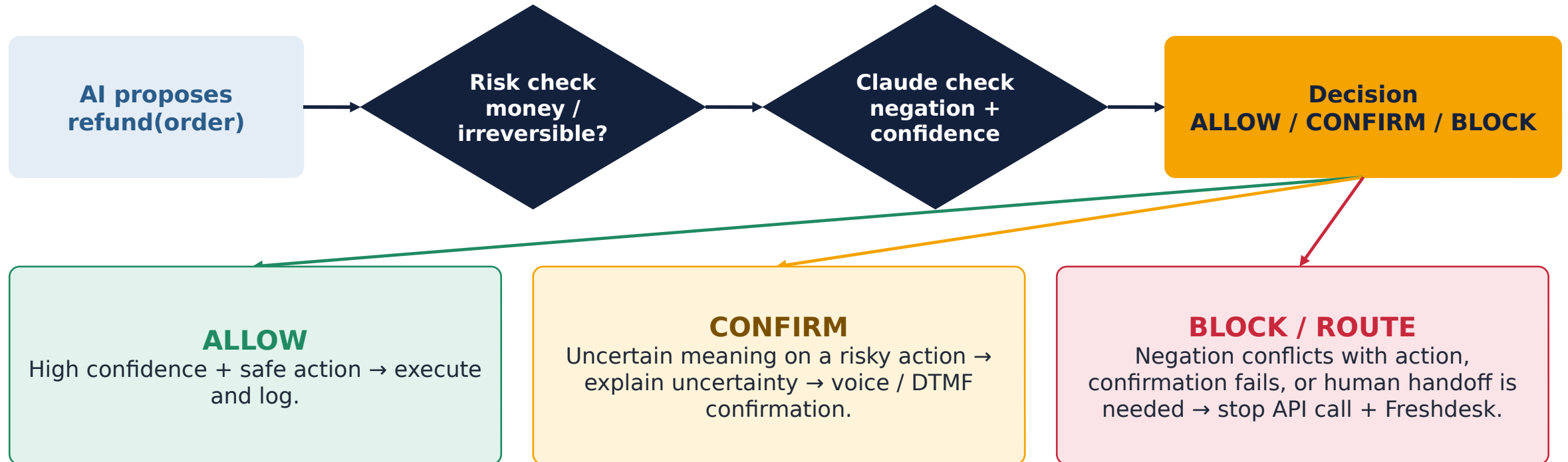
Every feature appears only where it adds value



Core principle Most routine calls take the fast path. Confirmation and Freshworks handoff activate only when risk or uncertainty requires them.

The Safety Decision

The agent never silently guesses on an irreversible action



EXAMPLE

"Refund vendam, replacement anuppunga."

BLOCK refund

Confirm replacement

If needed: Freshdesk ticket

MVP Scope & What Is Already Built

Separate what is demonstrated today from the intended MVP and future extension

DEMONSTRATED TODAY

- Round-1 code-mix tagging engine
- Intent scoring
- MCP server
- Live Freshdesk ticket API
- Vercel demo
- Freshdesk ticket creation flow

MVP SCOPE

- VoBiz + Sarvam + ElevenLabs voice path
- Claude safety gate
- Dodo action path
- Freshdesk escalation
- Hindi + Tamil + English
- Transcript + audit trail

NEXT PHASE

- Sentiment handoff tuning
- Interrupt handling tuning
- SOP rule editor + audit reports
- More Indian languages
- Freshworks Marketplace app
- Plug-in via MCP / REST

Credibility rule Any latency, accuracy or adoption number that has not been measured in the demo is presented as a target, not a result.

Target Customers & Value Proposition

Start where AI agents already have permission to take customer actions



Primary customers

D2C, e-commerce and fintech brands using Freshdesk and deploying AI voice agents for Indian customers.



Secondary customers

BPOs and support outsourcers that need one safety / policy layer across multiple client accounts.



Protected actions

Refunds • returns • replacements • cancellations • other company APIs exposed to the AI agent.

VALUE PROPOSITION

Plug it in once:

the existing AI agent can handle mixed-language customers while Codemix Skill...

- 1 validates intent
- 2 checks risky actions
- 3 confirms when uncertain
- 4 routes people to Freshworks when needed
- 5 records the full audit trail

14 Business Model, Go-to-Market & Resources

A simple commercial path: pilot → measurable proof → Freshworks distribution

PRICING

Free

monitor mode

\$5

per 1,000 guarded actions

Custom

Enterprise annual contract

Proposed pricing — validate in pilot.

GO-TO-MARKET

- 1 3 Indian D2C design partners
- 2 Free monitor-mode pilot
- 3 Publish wrong-action proof
- 4 Freshworks Marketplace listing
- 5 Scale through Freshdesk + BPO partners

COSTS & RESOURCES

~\$90K

proposed MVP budget · 3-month timeline



3 engineers



1 linguist / data annotator

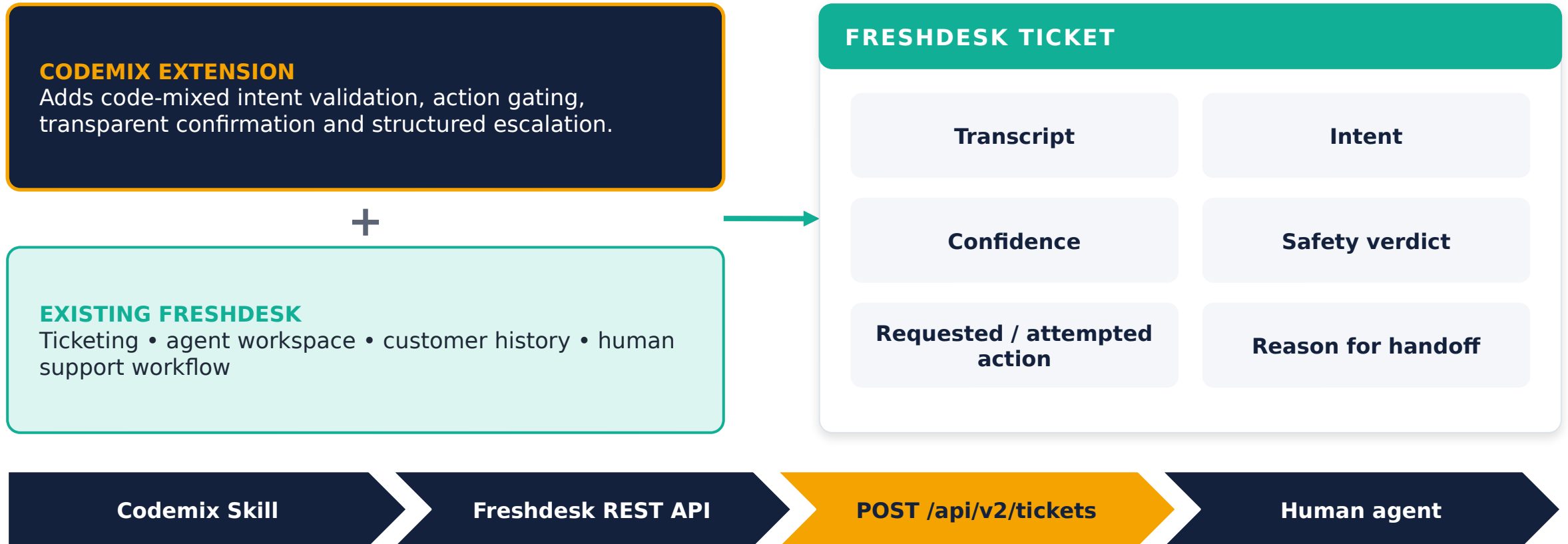


Part-time PM

Why Freshworks matters Freshdesk converts an unsafe/uncertain AI decision into an actionable, context-rich human support case.

15 Freshworks Integration — What We Add

Freshdesk becomes the human-in-the-loop destination when Codemix Skill decides the AI should not act alone



Positioning We extend Freshworks around the AI-action boundary; we do not replace Freshdesk or the company's existing APIs.

Freshdesk Ticket Payload

The handoff preserves the evidence a human needs to continue the conversation

Freshdesk ticket · created by Codemix Skill · POST /api/v2/tickets



Conversation context

Customer / call ID

Original transcript

Detected language / code-mix

English summary

Timestamp

Voice / DTMF source



Decision context

Detected intent

Confidence

Negation detected

Safety verdict

Requested API action

Confirmation status



Operational context

Priority / tags

SOP result

Reason for escalation

Attempted API call

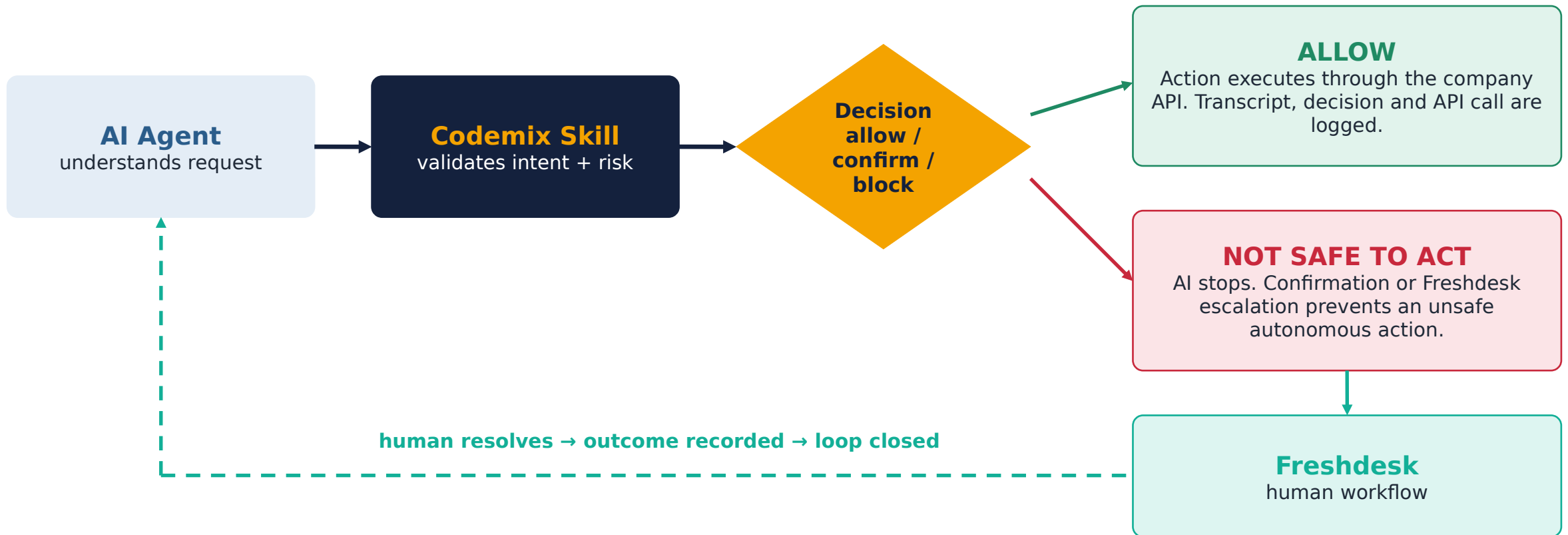
Human handoff reason

Audit reference

Judge takeaway Freshdesk is the system where uncertainty becomes a human workflow, with the evidence needed to act quickly.

Why Freshworks Is Central

Freshdesk is part of the control loop: safe actions continue automatically; uncertain or unsafe cases become human work



Freshworks value It closes the loop between autonomous action and accountable human support.

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Agentic Integration — The Plugin Control Point

Codemix Skill is middleware, not a replacement AI agent. It intercepts the proposed action before the company API is called

1 EXISTING AI AGENT

Understands the conversation and proposes a tool/API action.

2 CODEMIX SKILL

Receives transcript + proposed action → Claude checks intent and negation; the gate checks risk, SOP and confidence.

intercept

3 DECISION GATE

ALLOW → API call · CONFIRM → customer confirmation · BLOCK → stop action + Freshdesk handoff

4 COMPANY SYSTEMS

Existing APIs execute only after the gate. Freshdesk receives structured human handoffs.

Integration interfaces

MCP / REST adapter around the agent's tool calls. The company keeps its existing agent and business APIs.

Why this matters

One safety + policy layer can sit in front of multiple action APIs without rebuilding the voice agent.

Core positioning Plugin between the AI agent and its action tools.

Voice & Conversation Intelligence

The voice experience is designed around how Indian customers actually speak



Code-mixing

“En account la money varala.” Tamil + English can switch inside a single sentence. Sarvam → transcript → intent layer.



Negation

“Refund vendam, replacement anuppunga.” The presence of “refund” must not trigger a refund when “vendam” reverses it.



Interrupts

Caller can interrupt while the AI speaks. The agent stops, listens and treats the new utterance as the latest input.



DTMF / IVR

For exact confirmation, keypad input provides a reliable alternative to speech.



Sentiment

Angry / distressed caller → human handoff through Freshdesk.

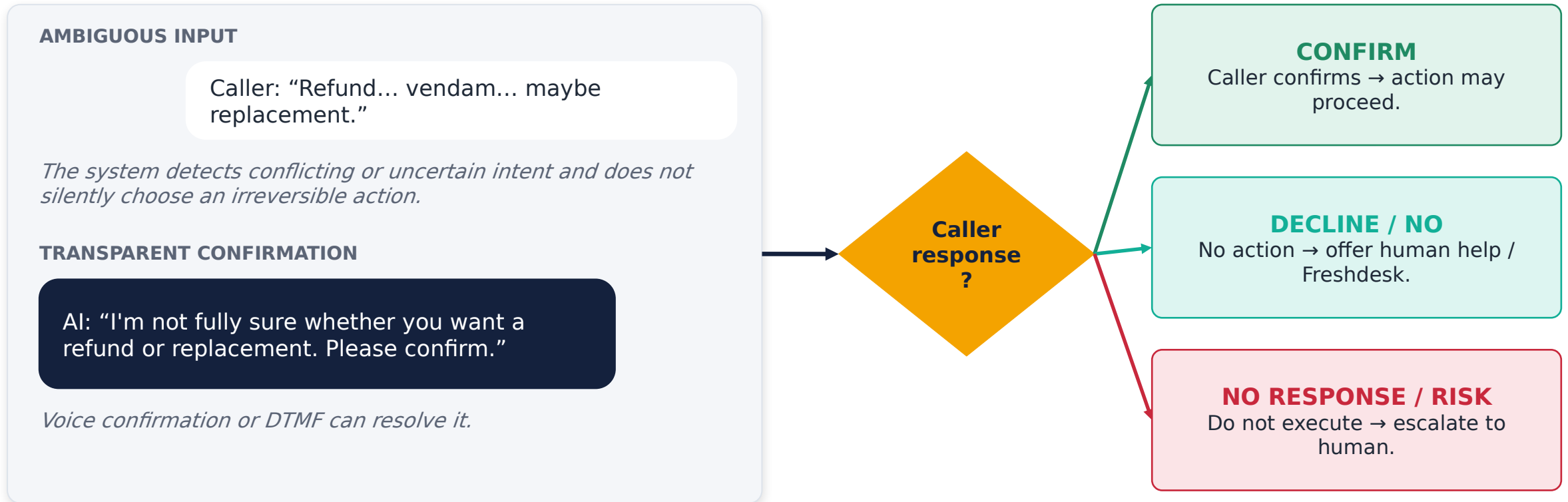


Natural response

ElevenLabs speaks the validated response and confirmation back to the customer.

Transparent Uncertainty

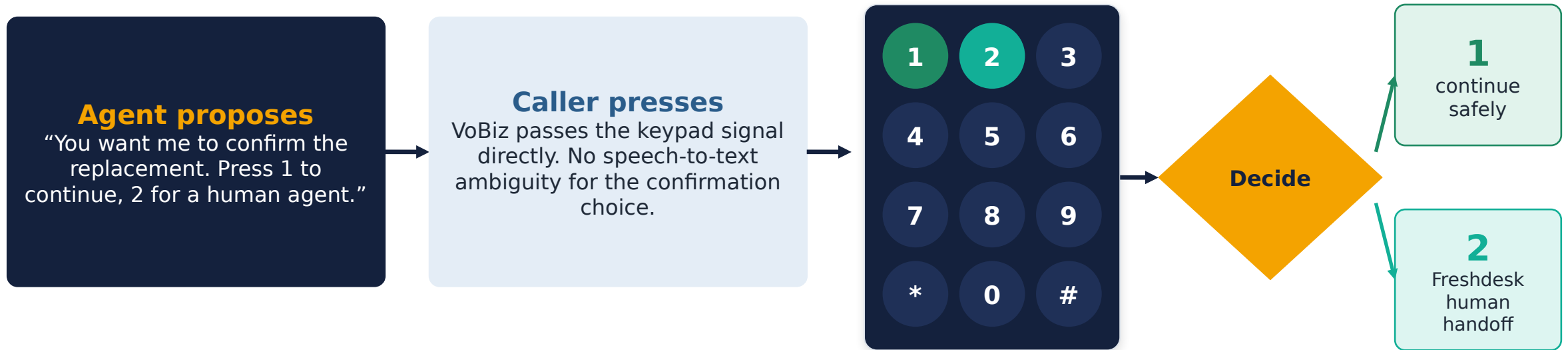
Instead of pretending to understand, the agent exposes uncertainty before a risky action



Principle Uncertainty becomes a visible decision point, not a hidden model guess.

DTMF + IVR Safety Path

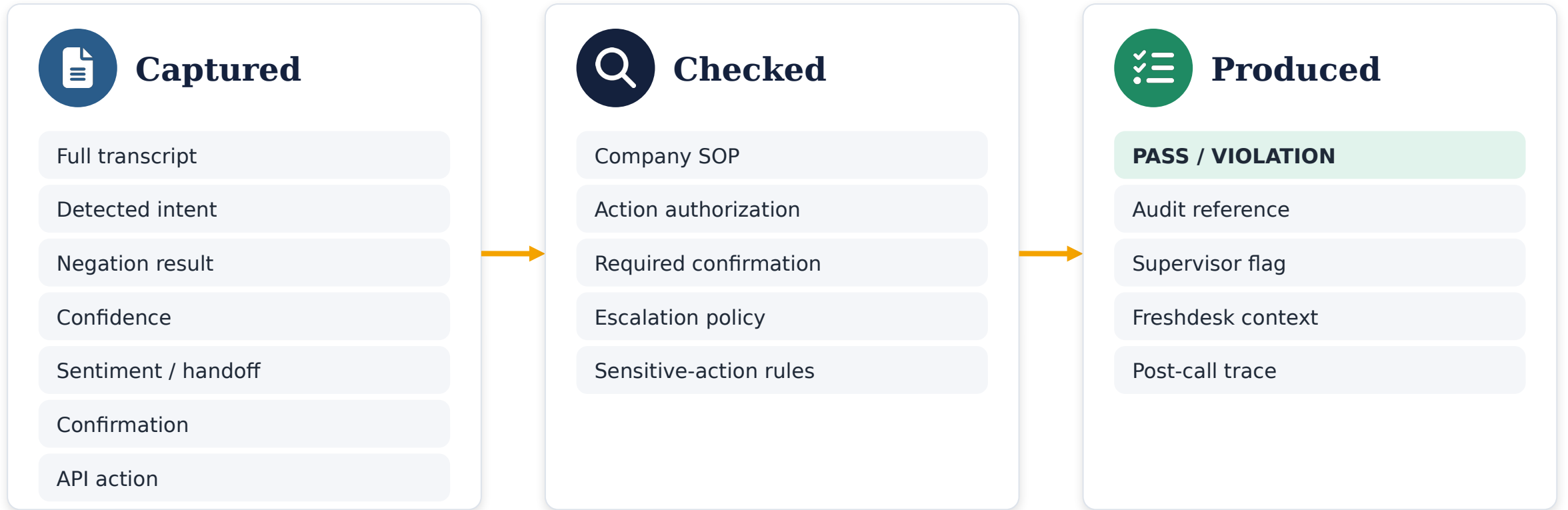
Numeric confirmation does not need speech recognition



Why it matters Order IDs, account numbers and binary confirmation choices are better represented by DTMF than by speech recognition. This gives the safety layer a deterministic input channel.

SOP-Based Auditing & Full Audit Trail

Every sensitive action can be reconstructed after the call

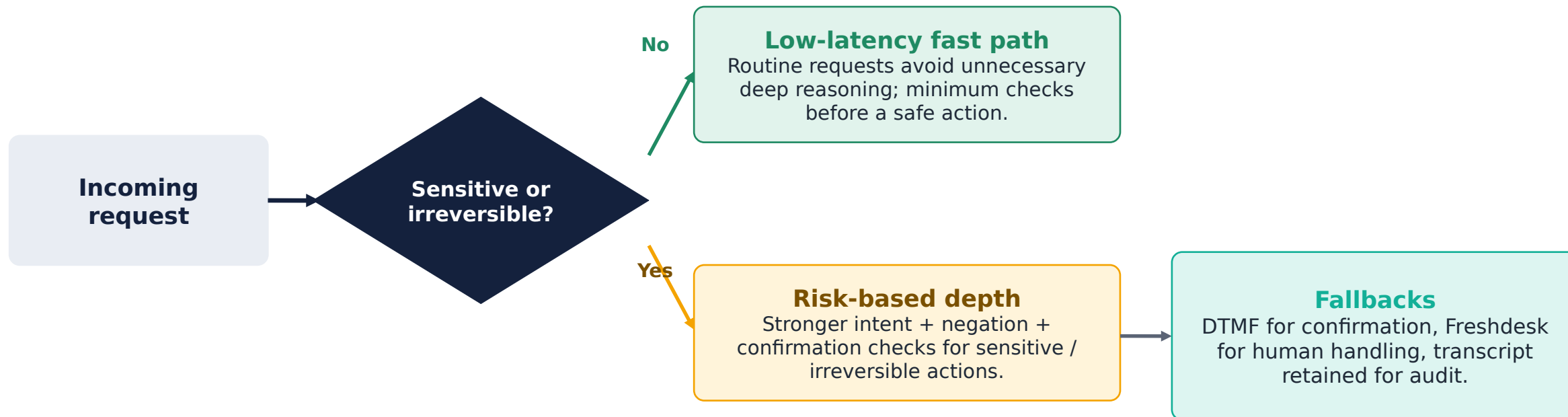


EXAMPLE AUDIT CHAIN



Performance & Reliability Design

Performance is a product feature, but measured claims are presented as targets unless already benchmarked



PROPOSED TARGETS (validate during pilot)

< 900 ms

p95 response target, sensitive-action path

≤ 15%

confirmation rate

100%

audit coverage

Success Metrics, Risks & The Ask

Measure safer actions, faster handling and complete accountability

< 0.5%

wrong actions

< 900 ms

p95 reply target

≤ 15%

confirmation rate

100%

audit coverage

-30%

reversal tickets

15%

adoption target

KEY RISKS → MITIGATION

STT drops negation

confirm risky actions

Latency

deeper checks only when needed

Dialect variation

lexicon + model fallback

Read-back friction

validate in pilot

OUR ASK

**Freshdesk sandbox +
Marketplace listing path for a
3-brand pilot.**

Use the pilot to measure wrong-action reduction, confirmation rate, latency and adoption.

Decision principle Do not silently guess on a sensitive action; expose uncertainty, confirm or route to a human.

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Final Pitch — Why This Is a Freshworks Extension

A concise closing story for the judges

THE PROBLEM

Code-mixed voice + negation can cause an AI agent to misunderstand an action.

THE PRODUCT

An intent-validation safety gate between what the voice agent hears and what it is allowed to do:
Understand → Decide → Act → Audit.

THE FRESHWORKS VALUE

When the AI should not act, Freshdesk turns uncertainty into a structured human workflow with transcript + decision context.

THE DEMO



“Don't cancel” should mean don't cancel — and Freshworks should know exactly what happened when the AI chooses not to act.

Codemix Skill • Understand → Decide → Act → Audit • Thank you